

HEMACHANDRU D

Coimbatore, Tamil Nadu, India | 8220404123 | hema007chandru@gmail.com | www.linkedin.com/in/hemachandru-deva-24451a290

PROFESSIONAL SUMMARY

Motivated and detail-oriented **Mechanical Engineering student** with a strong foundation in **manufacturing, machine design, engineering mechanics, thermodynamics, and industrial maintenance**. Hands-on experience through industrial training and academic projects in **mechanical design, automation, and IoT**, complemented by proficiency in CAD tools and engineering analysis. Passionate about solving real-world engineering challenges, optimizing processes, and contributing to continuous improvement initiatives.

EDUCATION

Sri Krishna College of Technology — B.E., Mechanical Engineering

CGPA: 9.08 Year : 2023 - 27

Government Model Higher Secondary School — HSC

Score: 85.5% Year : 2022 - 23

TECHNICAL SKILLS

CAD & Design Tools: SolidWorks, Fusion 360, AutoCAD.

Simulation Tools: ANSYS Workbench (Basic), COMSOL Multiphysics (Basic).

Programming Languages: Python, Java, C++.

Productivity Tools: Microsoft Excel, Microsoft Word, Microsoft PowerPoint, Microsoft Power BI (Basic), FIGMA(UI), Adobe Express.

AI Productivity Tools: ChatGPT, Gemini, Claude, Notebook LM, Lovable.

Professional Skills: Problem Solving, Analytical Thinking, Team Collaboration, Leadership, Design Thinking, Time Management, Adaptability, Communication.

INDUSTRIAL INTERNSHIP

NLC India Limited — SME–RC, Mine–I, Neyveli

28-Day Internship

- Studied the design, operation, and maintenance of industrial gearboxes used in Bucket Wheel Excavators (BWE), conveyors, and spreaders.
- Observed gearbox dismantling, inspection, reconditioning, welding, hard-facing, and bearing replacement processes.
- Analyzed common gearbox failure modes, including pitting, spalling, scuffing, and bearing fatigue.
- Gained knowledge of preventive maintenance practices, centralized lubrication systems, and condition monitoring techniques such as vibration analysis, oil analysis, and temperature monitoring to improve equipment reliability and minimize downtime.

PROJECTS

Smart Sewage Line Blockage Detection System Using IoT | Mini Project

- Developed an IoT-based system to detect sewage pipeline blockages using flow sensors and ultrasonic sensors.
- Integrated ESP8266 NodeMCU for real-time monitoring and cloud-based data transmission.
- Designed an early warning mechanism to improve maintenance efficiency and reduce sewage overflow.
- Tools & Technologies: ESP8266, Arduino IDE, IoT, Embedded Systems.

Design of an Integrated Machinery Setup for Chicken Waste Processing | Design Project

- Designed an integrated machinery system for processing chicken waste into value-added by-products.
- Applied machine design principles to improve process efficiency, hygiene, and operational safety.
- Developed a conceptual layout considering material flow, waste handling, and sustainable engineering practices.

IoT-Based Greywater Treatment and Monitoring System | Design Project

- Designed a smart greywater treatment system integrated with IoT-based monitoring for efficient water reuse.
- Proposed sensor-based monitoring to assess water quality and support sustainable water management.
- Focused on improving resource utilization through automation and real-time monitoring.

CERTIFICATIONS

- **IBM (Coursera)** – Introduction to Artificial Intelligence
- **Duke University (Coursera)** – Drones for Electric Vehicles
- **NPTEL** – Fundamentals of Additive Manufacturing Technology
- **NPTEL** – Computer Integrated Manufacturing
- **NPTEL** – Effective Writing
- **Cisco Networking Academy** – Introduction to Cybersecurity
- **Infosys Springboard** – Time Management
- **Infosys Springboard** – Mathematics.

IN-PLANT TRAINING & TECHNICAL ACTIVITIES

- **Ambal Autos** – Completed a one-day in-plant training, gaining practical exposure to automobile servicing, diagnostics, and maintenance operations.
- **Effica Automation Limited** – Completed a one-day in-plant training, learning the fundamentals of industrial automation, PLC control panels, sensors, and automated manufacturing processes.
- **EKVC 2025 (Electric/Engine Kart Vehicle Championship)** – Participated as a member of **Team Spartans**, representing Sri Krishna College of Technology in a national-level kart vehicle competition.

PUBLICATIONS & CONFERENCE PRESENTATIONS

- Presented the research paper "**Enhancing Mechanical and Functional Properties of Structural Composites Using Carbon Nanotube Reinforcement**" as an **Oral Presentation** at a conference organized by **Arashi Publications**.

TECHNICAL WRITING

- Authored the technical article "**Unveiling the Green Revolution in Casting: Harnessing the Power of Natural Binders**", featured in the **souvenir** of the **72nd Indian Foundry Congress** organized by the **Institute of Indian Foundrymen (IIF)**.

ACHIEVEMENTS

- Awarded **Second Rank** in the Department of Mechanical Engineering for outstanding academic performance during the **2024–2025** academic year.
- Awarded **Second Rank** in the Department of Mechanical Engineering for outstanding academic performance during the **2025–2026** academic year.

LANGUAGES

- **Tamil** – Native Proficiency
- **English** – Professional Working Proficiency